Introduction to calculus

2025/2026, Spring Trimester

Course faculty	Oleksii Horchakov e-mail: ohorchakov@kse.org.ua
Course assistants	Olga Prokopchuk e-mail: olechkadeadin@ukr.net Daria Slota e-mail: dslota@kse.org.ua
Department	Computer science
Study program	AI, SE, CS
ECTS credits	6 (180 hours)
Class hours	108 hours (15 lectures and 39 practice sessions)
Course language	Ukrainian, English
Course format	Offline
Form of study	Full-time studies

Prerequisites

«Introduction to mathematics» : functions, sequences, equations, inequalities, trigonometry

Background and course rationale

This course introduces its students to fundamental concepts in calculus, thus helping them further build their expertise in application of mathematical techniques. The course covers limits of sequences and functions and their properties, continuity, derivatives, integrals, differentiation and integration techniques, and basics of theory of series.

Learning outcomes

By the end of the course, students will be able to

- define the concept of limit of a sequence and a function, compute one-sided and two-sided limits of functions;
- apply various techniques for finding limits and recognize their applications in different contexts;
- compute the limits involving infinity, in particular, limits at infinity;
- analyze one-sided limits, continuous functions, and the properties associated with continuous functions;
- define the concept of derivative, distinguish between differentiability and continuity, recognize the practical meaning of derivative;
- apply basic rules of differentiation, including the chain rule, product rule, and quotient rule;
- calculate average rates of change and slopes of tangent lines, and interpret these in real-world contexts;
- calculate higher order derivatives;
- apply L'Hopital's Rule to evaluate limits of indeterminate forms;
- use derivatives to analyze monotonicity, convexity, and concavity of functions;

- find local extrema of functions using first- and second-order conditions, solve optimization problems using derivatives;
- use Taylor Polynomials to approximate functions;
- define the concepts of antiderivative and indefinite integral;
- use basic integration rules and the method of substitution to find indefinite integrals;
- evaluate definite integrals;
- find average values and determine areas between curves using definite integrals;
- define and apply the concept of improper integrals;
- to investigate the convergence or divergence of numerical sequences and series;
- represent functions in the form of infinite power series;

Methods of Teaching and Assessment

- **Lectures**: 15 offline lectures .
- **Practice sessions**: 42 offline sessions focused on problem-solving and active discussions.
- **Short quizzes**: 8 randomly assigned in-class quizzes.
- **Tests**: 3 in-class paper-based tests. Students are allowed to bring one handwritten A4 theoretical summary sheet.
- **Homework assignments**: 3 structured assignments submitted in PDF format, along with entering the results into a separate form in Moodle, which will evaluate the completed work.

Course Structure

This 15-week course consists of 42 classes of problem-solving and topic discussion, plus 15 lectures. The studying is conducted offline.

Tests. Students will take 3 in-class paper-based tests (16 points each). Students are allowed to bring one handwritten A4 theoretical summary sheet without examples and use it during the test. The summary must be submitted together with the student's work. No other sources may be used during the tests. If a student misses a test (worth 16 points) for a valid reason and informs the lecturer in advance, one common make-up session will be arranged for all such students. No retakes are available for any graded activities.

Quizzes. There will be 6 short in-class quizzes (3 points each), randomly assigned during the course. A student cannot write or rewrite a quiz later even if it was missed.

Homework Assignments. There will be 3 homework assignments (8 points each) submitted in PDF format. Deadlines and submission details will be listed in Moodle.

Communication. Course announcements, materials, and grades will be communicated via Moodle or Slack. It's the student's responsibility to regularly check the course Moodle page and their notifications for updates

Grading Policy

Written assignments (homeworks, quizzes and tests) grading. Taking into account the fact that the course is about mathematics and studying at the university involves observing the principle of scientificity, all answers and conclusions should be justified. In addition, a detailed explanation of the process of solving the problem is a good practice that allows a student to appeal in case he/she does not agree with evaluation of his/her work. Solutions or answers without sufficient explanation may receive a reduced or zero grade.

Homework

Homework problems and due time are announced on the course page in Moodle. All homework solutions should be written legible using dark ink on a white paper, photographed straightly without rotation, combined in order to a single pdf file, and submitted to Moodle before the announced deadline. Filename should contain first and last name of executant. Submissions not complying with all rules above will not be accepted or graded at the teacher's discretion. In this case, the student receives zero point for the assignment (no resubmission allowed).

We will have a late day policy on homeworks. Each student has one late day, i.e., you may submit your homework up to 24 hours after the deadline. Always reach out if you're facing unusual disruptions to your classwork.

You are allowed to work on the homework in small groups, but you must write up your own homework to hand in. Do not copy solutions from people you have worked with or from anyone else. If the answer is given in a book, don't just copy it, explain how you got it. Always indicate if you used a calculator for calculations.

Grade of the course is calculated as a sum of all the activities from the table above but not more than 100 points.

Task	***Number of activities***	***Point per activity***	***The total numbers of points***
Test	***3***	***18***	***54***
Homework	***3***	***8***	***24***
Quiz	***8***	***3***	***24***
Total			***102***

Academic integrity warning

Academic integrity is highly valued by the KSE community and its vast majority of students. We have a zero-tolerance policy towards academic plagiarism, self-plagiarism, fabrication, falsification, cheating, deception, bribery and other types of academic integrity violations. Due to the highly competitive nature of the programmes, all students must be treated equally. Even a single case of violation of academic integrity is a serious misdemeanour that may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions, and other merit-based awards. Therefore, penalties may vary from receiving zero points or a negative grade for an assignment to expulsion of the student from KSE, depending on the severity of the case and might include additional consequences such as deprivation of scholarships, financial assistance etc. All the rules and procedures regarding academic integrity are *stated in the KSE Code of Academic Integrity. Before enrolling in this course, students must be aware of and abide by the KSE Code of Academic Integrity.*

AI Tools usage policy

All tasks in all assignments must be written by the students on their own. It is forbidden to use any online calculators, mobile applications, chatbots and other AI tools, the same as the results of other people's work. You may use AI tools to brainstorm ideas or to check your work. For example, you might ask an AI to explain a concept you’re struggling with, or to double-check a calculation.

Course Faculty

Oleksii Horchakov*, PhD student at the Institute of Mathematics of the National Academy of Sciences of Ukraine, Teacher at Kyiv School of Economics.*

Holds a Master's degree from NTUU "KPI", Faculty of Physics and Mathematics.

Master's thesis: "Generalized intermittency in the Lorenz system".

Research interests: dynamical systems, chaotic dynamics.

Topic 1. Limits

Week 1

Lecture 1:

Sequences. Examples. Limit of sequence. Properties of convergent sequences. Number e

Practice session 1

Practice session 2

Practice session 3

Week 2

Lecture 2:

Limit of a function at a point. Cauchy (ε–δ) definition and Heine (sequence) definition. Properties of limits. One-sided limits.

Practice session 4

Practice session 5

Practice session 6

Week 3

Lecture 3:

Big-O and little-o notation. Special limits

Practice session 7

Practice session 8

Practice session 9

Week 4

Lecture 4:

Properties of functions continuous on a closed interval.

Cauchy’s intermediate value theorem. Weierstrass’s theorem.

Practice session 10

Practice session 11

Practice session 12

Topic 2. Derivatives

Week 5

Topic test. Limits

Lecture 5

Definition of the derivative. Motivation. Differentiation rules. Properties.

Practice session 13

Practice session 14

Practice session 15

Week 6

Lecture 6

Differentiable functions. Their properties. The theorems of Rolle, Lagrange, and Cauchy.

Practice session 16

Practice session 17

Week 7

Lecture 7

Differential of a function. Higher-order derivatives.

Practice session 18

Practice session 19

Week 8

Lecture 8

Taylor’s formula. L’Hôpital’s rule.

Practice session 20

Practice session 21

Practice session 22

Week 9

Lecture 9:

Convexity. The second derivative test. Graphing functions.

Practice session 23

Practice session 24

Practice session 25

Topic 3. Integral

Week 10

Topic test. Derivatives

Lecture 10

Antiderivative and indefinite integral. Substitution.

Practice session 26

Practice session 27

Practice session 28

Week 11

Lecture 11

Integrating by parts

Practice session 29

Practice session 30

Week 12

Lecture 12

Definite integral. Properties of the definite integral. Mean value theorem for integrals. Newton–Leibniz formula.

Practice session 31

Practice session 32

Practice session 33

Week 13

Lecture 13

Formulas of substitution and integration by parts. Some applications of the definite integral. Improper integrals.

Practice session 34

Practice session 35

Practice session 36

Topic 4. Series

Week 14

Lecture 14

Series. Convergence of series. Examples. Tests for convergence of numerical series.

Practice session 38

Practice session 39

Week 15

Lecture 15

Power series. Taylor series. Maclaurin series. Domains of convergence.

Practice session 40

Practice session 41

Practice session 42

Topic test. Integrals and series